

Key Findings

- **Paid media explains a meaningful share of subscriptions (40%-50%)** across model specifications, although baseline remains dominant.
- **Google is the primary incremental driver by scale, largely due to its spend share.** The conservative model structure limits over-attribution, producing posterior estimates consistent with diminishing marginal returns at current spend levels.
- **Several lower-spend channels appear efficient but are weakly identified.** Moloco shows favorable point estimates with wide credible intervals, while Beehiiv and LiveIntent exhibit multicollinearity. Aggregating Beehiiv, LiveIntent, and Amazon improves stability by reducing variance.
- **Subscriptions show a slow-moving seasonal pattern and short-term spikes in mid-July.** Paid media executions cannot plausibly explain the mid-July spikes.
- **Results are directional, not prescriptive.** ROI estimates and optimization outputs are prior-sensitive and best used to inform future testing rather than immediate decisions.

Project Description

I used MonthlyMocha's historical data to build a Marketing Mix Model with Google's Meridian package. The goal of the model is to estimate the incremental impact of paid media on subscriptions, and to quantify the uncertainty around those estimates.

My analysis is subject to several constraints:

- Approximately 1.5 years of weekly, national-level data
- No macroeconomic, promotional, or explicit seasonality controls
- A "first-shot" MMM intended to establish a baseline for future iterations

These constraints shape both the modeling strategy and the interpretation of results.

General Approach

KPI selection: subscriptions vs. revenue

I modeled subscriptions rather than revenue to anchor the analysis on an immediate, time-stamped outcome. A subscription occurs at a discrete time, making it easier to tie to recent media exposure.

In contrast, revenue accrues over the lifespan of a subscription and depends on downstream factors like retention. Modeling revenue directly would therefore introduce additional assumptions and obscure causal inference.

By modeling subscriptions, my analysis improves causal inference while still preserving the ability to translate results into revenue and ROI.

Modeling philosophy

Given the short time horizon and lack of non-media controls, the model relies heavily on generalization and prior-driven regularization. I used priors to encode plausible contribution levels and stabilize estimation.

The objective is to generate stable, interpretable results that support directional decision-making, not to produce maximally precise estimates. Where trade-offs arise, the model takes a conservative approach.

Data Overview

Data Received

The dataset contains approximately 1.5 years of weekly, national-level data, including subscriptions, paid media spend and impressions by channel, and an estimated revenue per subscription. I did not receive geographic breakdowns or non-media control variables.

I used subscriptions as the primary KPI, and created a constant `revenue_per_subscription` variable to enable translation of subscription-level results into monetary terms.

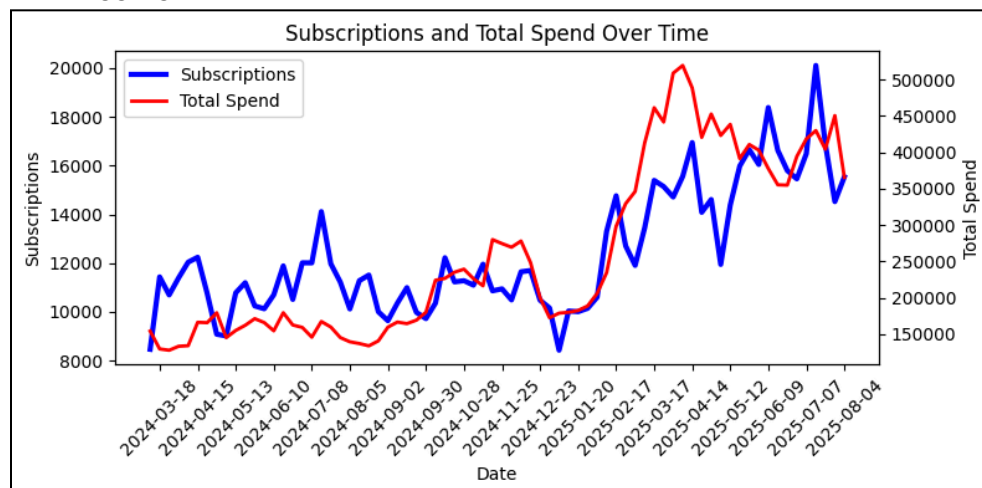
Structural Quality Checks

I ran basic structural quality checks to confirm that the data contained no missing values, duplicate rows, or duplicate dates. One channel (Roku) had zero spend and impressions throughout the entire period. Therefore, I removed Roku, since causal inference requires variation.

Exploratory Data Analysis

Total Spend v. Subscriptions

I first examined the relationship between total paid media spend and subscriptions to assess the extent to which aggregate demand moves with spend.



Overall, subscriptions track spend closely, suggesting that paid media contributes meaningfully to subscriptions. However, the relationship is not tight enough to imply that paid media alone explains the vast majority of demand. This indicates an important role for baseline and unexplained effects.

The plot also suggests a potential seasonal structure, with higher subscriptions relative to spend during warmer months, all else equal. This motivated me to test seasonal controls in the model.

Finally, two isolated increases in subscriptions appear in mid-July 2024 and 2025 that cannot be plausibly attributed to paid media activity. I noted these events as structural anomalies and handled them explicitly in the modeling stage to prevent spurious paid media attribution.

Correlations and Multicollinearity

I examined pairwise correlations between media impressions and subscriptions.

Variable 1	Variable 2	Correlation
subscriptions	beehiiv_impressions	0.849558
liveintent_impressions	beehiiv_impressions	0.700497
subscriptions	google_impressions	0.676619
google_impressions	beehiiv_impressions	0.644869
subscriptions	liveintent_impressions	0.625381

Google, Beehiiv, and LiveIntent exhibit strong correlations with subscriptions. Beehiiv also shows high correlation with both Google and LiveIntent, raising concerns about multicollinearity and weak independent identification.

VIF analysis confirms this issue. Beehiiv’s impressions have a VIF of 6.3, indicating a high level of multicollinearity that complicates channel-level attribution. This finding informed later decisions intended to reduce model complexity. No other channels have problematic VIFs.

Share of Total Spend by Channel

Paid media is highly concentrated, with Google accounting for over half of total paid media spend. Several channels, including Moloco Beehiiv, LiveIntent, and Amazon, individually account for less than 5% of total spend.

Low-spend channels provide limited signal and are more susceptible to collinearity effects, particularly in short-term data that lacks controls. The spend distribution further supports conservative treatment of these channels—especially Beehiiv, given its high VIF.

Key EDA Takeaways

- Paid media is strongly associated with subscriptions, but does not fully explain demand
- Subscriptions likely follow a seasonal pattern, with short-term spikes in mid-July
- Google, Beehiiv, and LiveIntent are highly correlated with subscriptions
- Beehiiv exhibits significant multicollinearity with other media channels
- Media spend is skewed across channels, with several low-spend (<5%) channels

Modeling Assumptions and Decisions

This section outlines the rationale for each major modeling decision. My decisions were guided by my goals of preserving causal interpretability under limited data and mitigating early signs of substantial overfitting. Each decision was stress-tested across multiple specifications to assess stability.

Channel Combinations

Early diagnostics showed that certain low-spend channels had high multicollinearity, weak independent signal, or both. After observing persistent train-test divergence even after adjusting priors, lags, and random effects, I evaluated whether channel-level separation was driving instability.

I grouped Beehiiv (high VIF), LiveIntent (Beehiiv's strongest covariate), and Amazon (lowest spend) into a single combined predictor ("Below Threshold") to reduce dimensionality and improve identifiability. This approach traded granularity for more stable attribution, thereby reducing overfitting.

Priors

Because subscriptions are a non-revenue KPI, I used contribution-based priors instead of ROI priors to maintain interpretability across both subscription and revenue contexts. I anchored prior means to each channel's share of total spend. A channel's prior contribution mean is 50% of its spend share, to reflect my estimate that non-media drivers explain 50% of subscriptions. I varied prior strength (k) to assess sensitivity.

I applied a baseline k of 70 to all channels except Google and Below Threshold (see reasoning below). These priors act as conservative regularization rather than hard constraints. Although their means sum to ~50%, they are applied independently, allowing baseline to explain a larger share of subscriptions if strongly supported by the data.

1. Google

Google impressions show a strong correlation with subscriptions, raising a risk of over-attribution of random noise. I increased k from 70 to 100 to require stronger evidence for Google to absorb variation beyond its expected contribution. This adjustment reduced the risk of overstating Google's contribution while preserving stable directional results.

2. Below Threshold

Below Threshold aggregates several tactics that exhibit either very low spend or high collinearity. Since this combined channel represents a mix of channels with weak individual identification, I increased its prior from 70 to 100 to reduce the risk of misattribution. This decision favors conservative, stable estimates over granularity.

The combined “Below Threshold” channel aggregates low-spend and highly correlated tactics. I increased its prior from 70 to 100 to treat this heterogeneous group conservatively. As with the Google adjustment, this choice favors conservative estimates over unreliable attribution.

Random Effects

Given the short time horizon and absence of non-media controls, significant unexplained variation is expected. I tested random effects contribution priors ranging from 3% to 10%. Higher priors allow non-media factors to absorb noise that would otherwise be misattributed to paid media. A 10% prior most effectively reduced overfitting and better aligned with my expectations, given the relatively sparse data.

Adstock Type

I modeled the adstock effect using a simple, interpretable structure: geometric effects. With limited data and controls, higher-degree adstock structures risk overfitting. A geometric adstock provides a simple, transparent decay assumption appropriate for a first-shot MMM.

Adstock v. Hill Sequencing

I used the default Adstock v. Hill sequencing for ease of interpretability. Media exposure accumulates over time through the Adstock effect. By applying the Hill function after Adstock effects, the model applies diminishing returns to accumulated media exposure, not to each week in isolation. This reduces the risk of saturation (Hill) fitting short-term noise.

Maximum Lags

Lag length directly affects attribution of baseline trends to media. I tested maximum lags ranging from one to four weeks. Shorter lags reduce the risk of over-attribution. I opted for a two-week maximum lag across all channels, primarily because it matched my expectations for MonthlyMocha’s business. It is unlikely that a social media impression would drive coffee shop subscriptions one month later. Lower maximum lags also improved out-of-sample stability.

Baseline Knots

I intentionally constrained baseline flexibility to avoid overfitting. I tested one to four knots under the assumption that fewer knots favor interpretability and prevent capturing noise. I found that two knots allowed for modest baseline variation without degrading generalization.

July Spike Dummy

Two isolated, one-week spikes in subscriptions occurred in mid-July 2024 and 2025 that could not plausibly be explained by paid media activity or seasonality. I introduced a dummy variable to absorb these spikes, thereby protecting baseline and media estimates. The primary purpose of the dummy was to preserve causal interpretability. As a side effect, the dummy improved in-sample and out-of-sample fit.

Test and Train Split

I used holdouts diagnostically rather than to forecast optimization, in line with Meridian's design. I tested holdout proportions ranging from 10% to 20%. Results were directionally consistent across these ranges, so I used a 10% split for my final model.

Markov Chain Monte Carlo Parameters

My goal in selecting MCMC parameters was to optimize convergence and posterior stability. I was more concerned with ensuring stable posterior inference than with prediction. I mostly maintained default settings, but increased `n_adapt` from 500 to 1000 to improve mixing. While this change slightly increased runtime, the final runtime was reasonable, at approximately 4 minutes. My final settings achieved stable inference with $R\text{-hat} < 1.01$ across parameters.

Final Model Specifications

The specifications below reflect the final model used for all reported results.

Model type

- Bayesian MMM
- Weekly national-level data

KPI

- Subscriptions (non-revenue KPI)

Media inputs

- Paid media impressions and spend by channel
- Low-spend / highly correlated channels aggregated into Below Threshold

Baseline and controls

- Smooth baseline with 2 knots
- Seasonal sine/cosine terms
- Mid-July spike indicator

Adstock and saturation

- Geometric adstock
- Hill function applied post-adstock
- Maximum lag: 2 weeks

Priors (see Jupyter Notebook for full distributions)

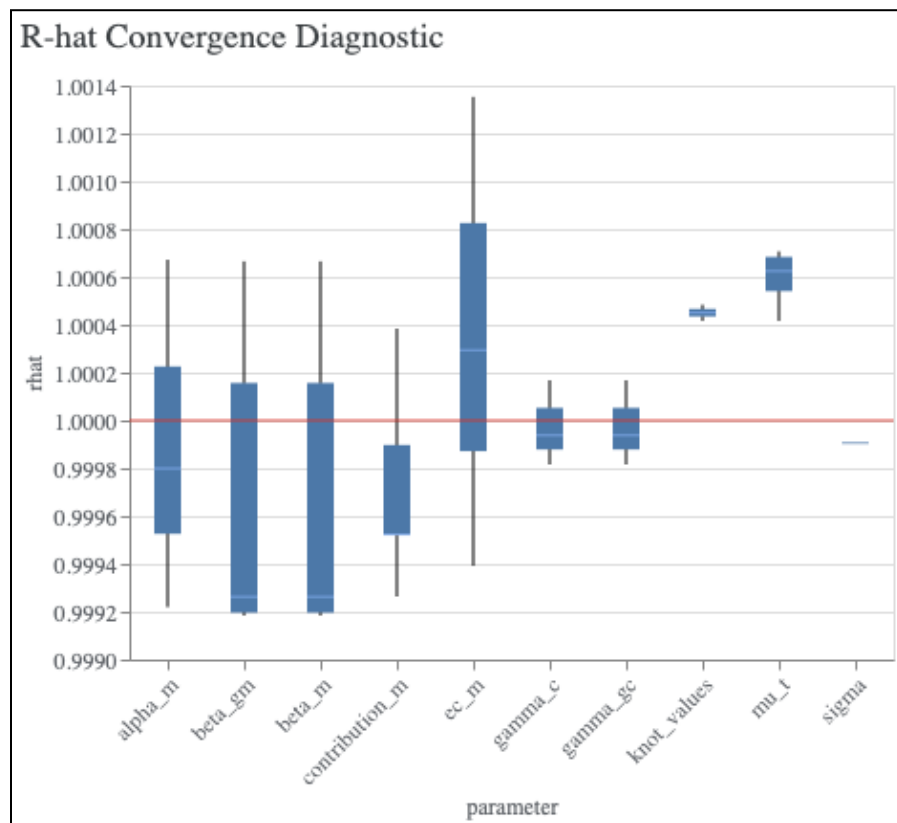
- Contribution priors centered at 50% aggregate media share
- Channel-level priors scaled by spend share
- Stronger priors applied to Google and Below Threshold
- Random effects prior mean: 10%

Estimation

- 7 MCMC chains
- $N_{\text{adapt}} = 1000$, $n_{\text{burnin}} = 500$, $n_{\text{keep}} = 1000$
- $R\text{-hat} < 1.01$

Model Diagnostics

Convergence

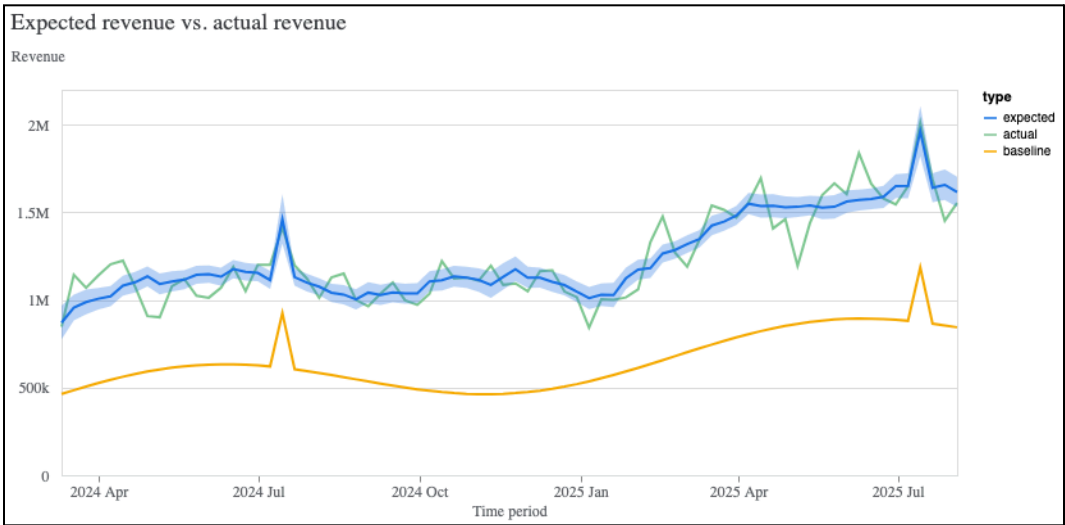


All monitored parameters converged well, with R-hat values below 1.001 across chains, indicating stable posteriors.

Model Fit Statistics

The model explains a substantial share of variation in subscriptions (Train R-Squared = 0.83, Test R-Squared = 0.67), with test error of approximately 8% (wMAPE). This level of fit is consistent with a causally-focused mode that prioritizes stability. The remaining test-train gap reflects the limited controls in the data.

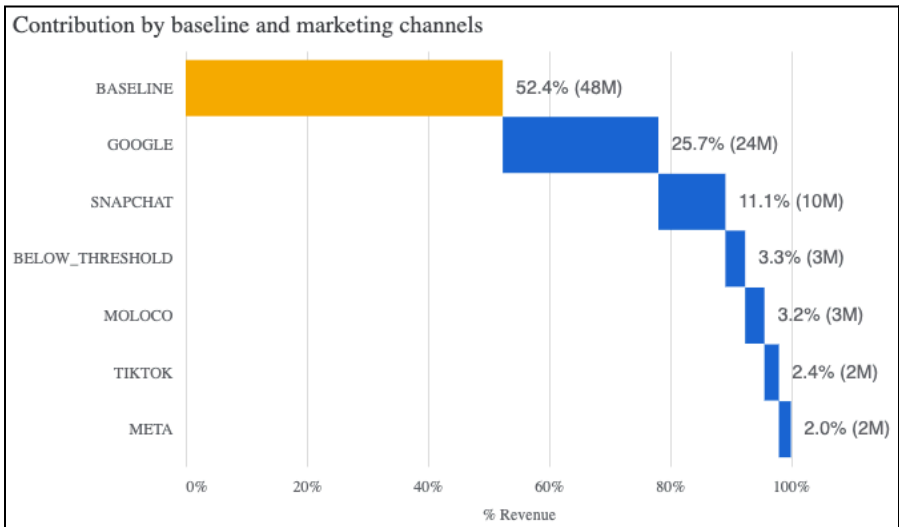
Expected Revenue vs. Actual Revenue



The model captures the overall subscription trend well. It also smooths short-term variation, which the model largely attributes to random effects rather than media.

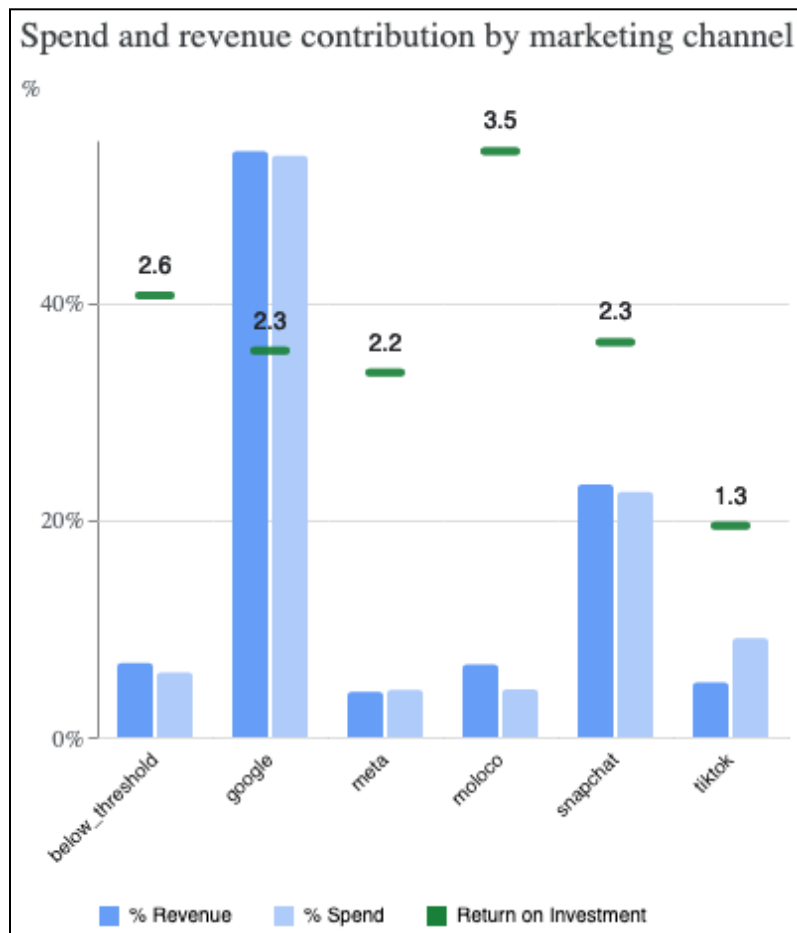
Media Findings

Overall Attribution and Baseline Role



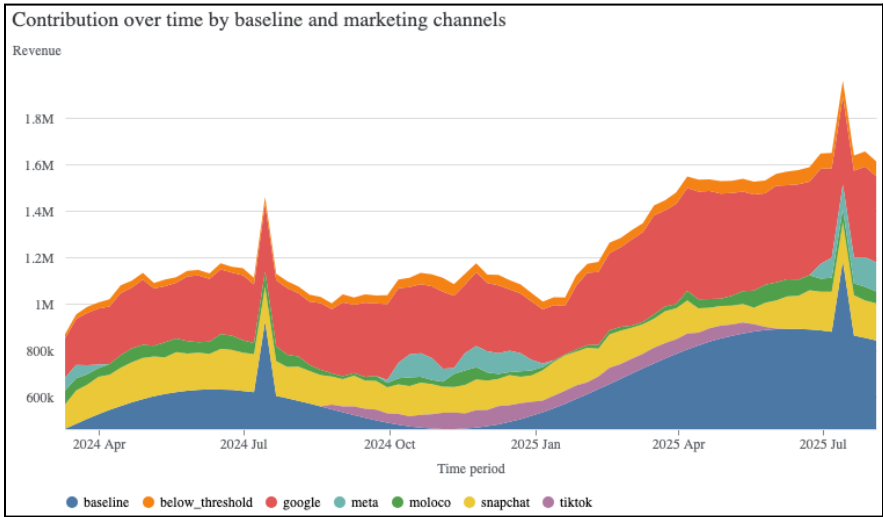
The waterfall chart shows that baseline demand pushed the prior distribution upward. But paid media still explained a meaningful share of subscriptions. Google is the largest incremental driver. This distribution is consistent with the conservative specification that stabilizes contributions around share of spend.

Spend and Revenue Contribution by Marketing Channel



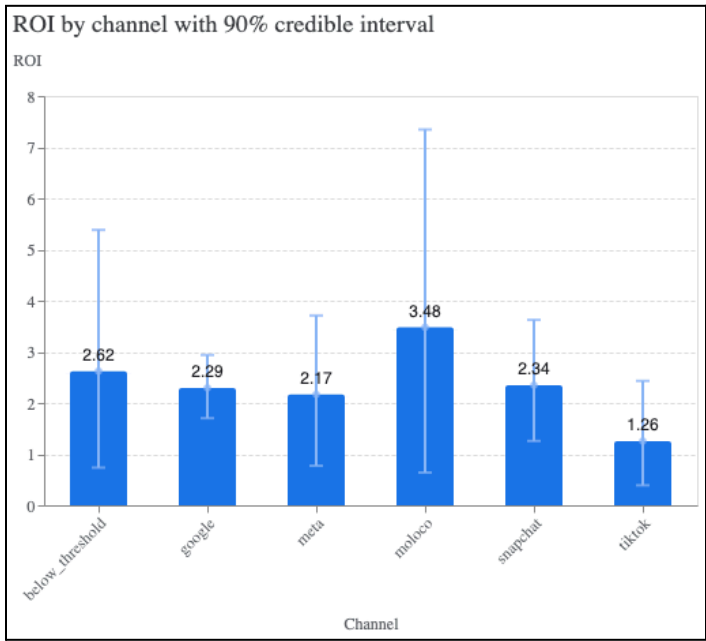
While heavily influenced by priors, the spend share to contribution comparison highlights meaningful efficiency across channels. Snapchat and Moloco appear to outperform their spend levels. However, lower-spend channels should be interpreted cautiously, given limited signal.

Contribution Over Time by Baseline and Marketing Channels



Channel-level contributions evolve slowly over time, rather than tracking weekly fluctuation. This behavior supports the model’s emphasis on generalizability.

ROI by Channel with 90% Credible Interval



Estimated ROI varies across channels and should be interpreted cautiously due to the limited time horizon and strong priors. While Moloco and Below Threshold show high point estimates, they also have wide confidence intervals. For this first-shot MMM, ROIs should be interpreted directionally, with minimal weight placed on rankings.

Taken together, these results suggest that paid media plays a meaningful role in driving subscriptions, with Google serving as the primary high-spend driver. The model produces stable attribution over time, consistent with spend dynamics. These findings are best suited for directional conclusions rather than immediate optimization decisions.

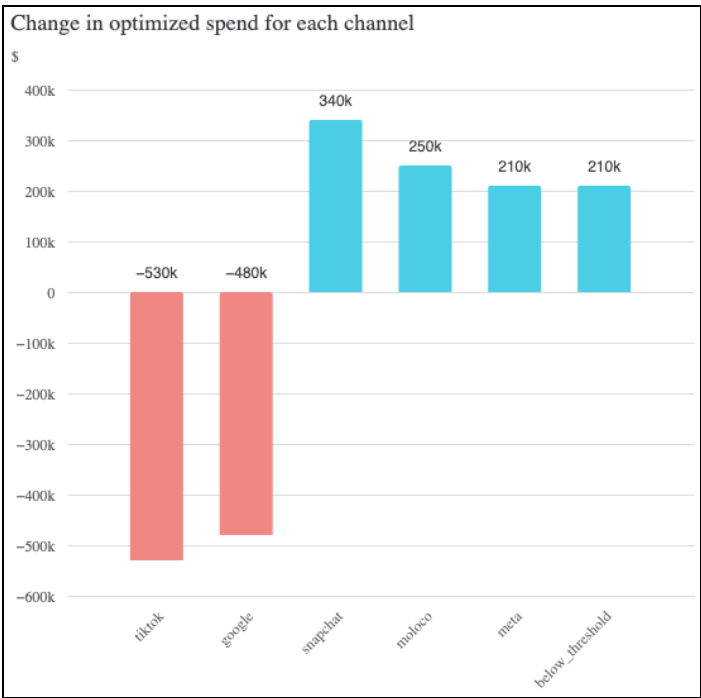
Budget Optimization

Purpose and Limitations

For illustrative purposes, I assume the objective is to maximize subscriptions on a fixed budget. As such, I used Meridian’s Optimize to KPI framework.

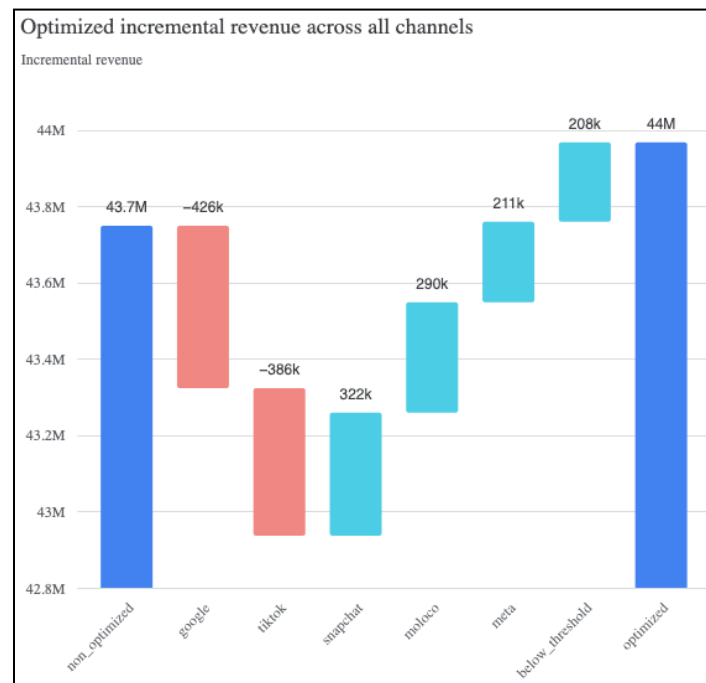
This optimization should be interpreted as a thought experiment rather than a prescriptive recommendation. The results are highly sensitive to contribution priors, among other assumptions and limitations.

Optimized Spend Reallocation by Channel



Google (\$530k) and TikTok (\$480k) spend is reallocated toward Snapchat, Moloco, Meta, and the combined channels.

Optimized Incremental Revenue by Channel



The optimized budget yields a roughly \$300,000 increase in incremental revenue from subscriptions. This increase is small relative to total revenue (\$43.7m), reinforcing the directional nature of the model.

Future Analysis

Based on this analysis, the following steps describe how this first-shot MMM can be improved to support more reliable measurement and optimization.

- **Extend the time period and add explicit controls.** The use of macroeconomic indicators, promotions, price, and holidays may reduce reliance on priors.
- **Increase channel identification through experimentation.** For example, geo-based lift experiments may improve identifiability for lower-spend and correlated channels.
- **De-aggregate combined channels as more data accumulates.** This would allow for efficiency comparisons between Beehiiv, LiveIntent, and Amazon.